

SalPurFlask (Sales Purchases & Inventory Management System)

This is a Flask-based web application designed for an inventory management system named "SalPurFlask." Its primary objective is to enable users to manage records of suppliers, customers, items, purchases, and sales efficiently. The application encompasses user authentication (signup, signin, email verification, password reset), data management (CRUD operations), and reporting functionalities.

Technologies Used

- **Flask:** A web framework used for routing, templating, and server handling.
- **Flask-SQLAlchemy:** Facilitates database operations with an SQLite database.
- **Flask-Login:** Manages user authentication and session handling.
- **passlib:** Implements password hashing using the Argon2 algorithm.
- **smtplib:** Enables email sending for verification and password reset.
- **itsdangerous:** Generates secure tokens for email verification and password reset.
- **Flask-Paginate:** Provides pagination for data tables.
- **python-dotenv:** Loads environment variables.
- **SQLAlchemy:** Offers Object-Relational Mapping (ORM) for database interactions.
- **CSV:** Generates CSV files for data export.

Application Features

1. **User Authentication:**
 - User signup with email verification.
 - Secure signin with password hashing.
 - Password reset via email.
 - Option to resend email verification.
 - User signout functionality.
2. **Inventory Management:**
 - Create, Read, Update, and Delete (CRUD) operations for suppliers, customers, and items.
 - Record and manage purchases and sales.
 - Stock management with automatic updates to item stock levels.
3. **Reports:**
 - Date-wise purchase and sale reports.
 - Reorder level reports for low-stock items.
 - Profit analysis by date, item, and customer.
 - Export functionality for all reports in CSV format.
4. **Security:**
 - Password hashing using Argon2 for secure storage.
 - Email verification to ensure secure access.
 - Session management with Flask-Login.

File Structure and Functionality

- **Models:** Defines database models for User, Supplier, Customer, Item, Purchase, and Sale.
- **Routes:** Implements routes for authentication, dashboard, CRUD operations, and reporting.
- **Utilities:** Includes helper functions for email sending, token generation, and verification.
- **Decorators:** Utilizes `@login_required` and `@verified_required` to secure routes.

File: app.py

Purpose: This is the main Flask application file that handles the core logic of the inventory system. It includes user authentication (signup, signin, etc.), supplier and customer management, and database operations. This file, defining routes for items, purchases, sales, and reports. It includes essential CRUD operations and reporting features for inventory management.